

Nirma University
Institute of Technology
Semester End Examination (IR), May - 2026
M. Tech. in CSE/M. Tech. in CSE (Data Science), Semester - II
6CS376ME25 - EXPLAINABLE AI

Roll /
Exam No. 25mcD005

Supervisor's initial
with date

A.S. 05/05/26

Time: 3 Hours

Max. Marks: 100

- Instructions:
1. All questions are compulsory. (No Optional Questions)
 2. Use section-wise separate answer book.
 3. Figures to right indicate full marks.
 4. Assume necessary data wherever required and specify them.
 5. Avoid attempting questions and their sub-questions in a random order.

Section - I

- Q-1.** Answer following questions: [20]
- (A)** A machine learning model is designed such that its internal working can be directly understood, while another requires separate methods to interpret its decisions after training. Explain the difference between intrinsic and post-hoc explainability, and discuss how model-agnostic explanations differ from model-specific ones in terms of applicability and flexibility. [5]
- CO1,
BL2
- (B)** Consider a machine learning model defined as $F(x_1, x_2) = x_1 * x_1 + 2 * x_2$, where $x = (2, 3)$ is the input instance and $x' = (0, 0)$ is the baseline. Using the concept of Integrated Gradients, answer the following: [15]
- CO4,
BL5
- i.) Define Integrated Gradients with its mathematical equations.
 - ii.) Explain the significance of choosing an appropriate baseline in attribution methods.
 - iii.) Compute the interpolated inputs for $k = 1, 2, 3, 4$ using Reimann Equation (approximated and discrete version).
 - iv.) Derive the expressions for the partial derivatives $\partial F / \partial x_1$ and $\partial F / \partial x_2$, and evaluate them at each interpolated point.
 - v.) Using the Riemann approximation of the integral, compute the Integrated Gradients for both features x_1 and x_2 .
 - vi.) Verify whether the completeness property holds, and comment on any approximation error observed.
 - vii.) Based on the computed attributions, interpret which feature contributes more to the model prediction and justify your answer.
- Q-2.** Answer following questions: [20]
- (A)** Analyse how temperature, humidity, and traffic density affect air pollution levels, where relationships are expected to be non-linear. Describe the concept of a Generalized Additive Model (GAM) and how
- CO2,
BL3
- [05]

(B)
CO2,
BL3

it extends GLM, write its general mathematical form, and explain why GAM is more suitable for this problem compared to GLM.

Consider the following regression model and the data:

[15]

$$f(x_1, x_2, x_3) = 3x_1x_2 + 2x_1x_3 + x_1$$

x_1	x_2	x_3
2	2	2
2	2	3
3	1	2
2	3	1
3	3	3

For the instance: $(x_1, x_2, x_3) = (2, 1, 2)$ perform the following:

- Write the standard Shapley value formula used to compute feature contributions along with the Shapley interaction value formula for features i and j and define all its components.
- Define main effect and interaction effect in SHAP. Explain how non-linear feature interactions influence predictions and how SHAP interaction values help identify such dependencies.
- Compute the baseline value of the model output.
- Compute all required intermediate model outputs including $f(\emptyset)$, $f(x_1)$, $f(x_2)$, $f(x_3)$, $f(x_1, x_2)$, $f(x_1, x_3)$, $f(x_2, x_3)$, and $f(x_1, x_2, x_3)$.
- Compute the SHAP interaction values for all feature pairs ϕ_{12} , ϕ_{13} , ϕ_{23} .
- Briefly interpret which features exhibit interaction and explain what a non-zero interaction value signifies.

Q-3.
CO3,
BL4

Explain the concept of Responsible AI and examine its key principles such as fairness, transparency, accountability, and privacy. How do these practices help in addressing the risks associated with AI systems in real-world applications?

[10]

Section – II

Q-4.
CO4,
BL5

For LIME explainable model consider $\sigma=2$ and assume a complex model, $f(p, q) = p \cdot 0.2 + q \cdot 0.5 + 0.7$ to answer following questions for an instance/sample $x = (p, q) = (10, 20)$. Consider Euclidean distance as a distance calculation method and following perturbed samples:

[20]

Perturbed Samples	p	q
z1	8	21
z2	9	19
z3	11	22

- Why is perturbed data required in LIME?
- Explain why LIME assigns lower weights to distant samples.
- Explain what would happen if all perturbed samples were given equal weight (i.e., no proximity kernel used).
- Write equation of an objective function with penalty and explain every component of it.
- What prediction a complex model takes for the sample x . Calculate value for the prediction.

- vi.) Calculate prediction of a complex model for the perturbed Samples.
 vii.) Compute $\pi_x(z_i)$ for each perturbed samples from x .
 viii.) Find the value(s) of weighted square loss.

- Q-5.** Answer following questions: [20]
(A) A trained CNN produces two feature maps $A^1, A^2 \in \mathbb{R}^{2 \times 2}$ for a given [10]
 CO1, input image, along with the corresponding gradients of a target class
 BL3 score y^c . Construct the complete system of equations required to
 obtain the final class-specific heatmap using Grad-CAM.
(B) A CNN produces feature maps: $A^1 = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ and $A^2 = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$. [10]
 CO4, Considering the gradients: $dY/dA^1 = \begin{bmatrix} -0.2 & 0.1 \\ 0.3 & -0.1 \end{bmatrix}$ and
 BL3 $dY/dA^2 = \begin{bmatrix} 0.4 & -0.3 \\ 0.2 & 0.1 \end{bmatrix}$, compute the final Grad-CAM heatmap
 (after applying ReLU and normalization) for the target class.
Q-6. Answer following questions: [10]
(A) Explain the concepts of Demographic Parity and Equal Opportunity. [5]
 CO3, Compare their assumptions and limitations, also assume a scenario
 BL4 and analyze where satisfying one metric may violate the other.
(B) Discuss the application of Explainable AI in the finance sector. [5]
 CO3, Evaluate how Explainable AI supports risk assessment and fraud
 BL4 detection, and examine the challenges of ensuring fairness, avoiding
 discriminatory outcomes, and maintaining transparency in AI-driven
 financial systems.